

| Question number | Answer                                                                                                                                                                                                             | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5(a)            | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• can differentiate into all cell types (1)</li> <li>• have no Hayflick limit / can divide indefinitely (1)</li> </ul> | (2)  |

| Question number | Answer                                                                  | Mark |
|-----------------|-------------------------------------------------------------------------|------|
| 5(b)(i)         | B negatively-charged fragments move towards the positive end of the gel | (1)  |

| Question number | Answer | Mark |
|-----------------|--------|------|
| 5(b)(ii)        | A 1    | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 5(b)(iii)       | <p>Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme.</p> <p>The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant.</p> <p>Indicative content</p> <ul style="list-style-type: none"> <li>• Bands 1, 2, 5, 9 and 11 in the embryo are also found in the mother</li> <li>• therefore, they must have been present in the nucleus of the egg cell from the mother</li> <li>• Bands 4, 6, 7, 8 and 10 in the embryo are also found in the father</li> <li>• therefore they must have been present in the nucleus of the sperm from the father</li> <li>• Bands 3 and 12 in the embryo are not found in either the mother or the father</li> <li>• therefore they must have been the mitochondrial DNA from the donor</li> <li>• both mother and father have bands not found in the embryo</li> <li>• the bands in the mother could be her mitochondrial DNA</li> </ul> | (6)  |

| Level | Marks | Descriptor                                                                                                                                                                                                                                                                                                                          |
|-------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | 0     | No awardable content.                                                                                                                                                                                                                                                                                                               |
| 1     | 1-2   | <p>An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information.</p> <p>The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.</p>       |
| 2     | 3-4   | <p>An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information.</p> <p>The explanation shows some linkages and lines of scientific reasoning, with some structure.</p>                                                                            |
| 3     | 5-6   | <p>An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information.</p> <p>The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.</p> |

